

Week 03: Diffusion and generated media

Generative AI | MSc teaching notes and further reading | 2027

Learning target

Explain the denoising idea and judge an image with a consistent checklist.

Context and teaching plan

A college needs a synthetic poster image while respecting consent and image rights.

Contact time	Suggested use
Two hours: concepts	Case introduction, four ideas, worked example and misconception.
Two hours: practice	10 min recall, 20 min demo, 40 min guided practice, 25 min independent change, 15 min review, 10 min exit check.

Explain the key ideas

Forward noising

In a common training setup, progressively add noise to a clean image.

Teacher question: Where does forward noising appear in the weekly case, and what could go wrong if we omit it?

Denoising model

A learned model estimates how to remove noise at each step; the course simulation uses the known clean target.

Teacher question: Where does denoising model appear in the weekly case, and what could go wrong if we omit it?

Conditioning

Text or other inputs can guide generation. The condition does not guarantee accurate details.

Teacher question: Where does conditioning appear in the weekly case, and what could go wrong if we omit it?

Image review

Check whether text, people, permissions and task-specific details meet the brief.

Teacher question: Where does image review appear in the weekly case, and what could go wrong if we omit it?

Worked example and answer guide

- Input: Clean grid has an L-shaped bright pattern.
- Change or question: Noise obscures the L at high strength.
- Supported observation: A toy restoration with access to the clean target restores it; a real model does not have that target.

Reasoning: state the given evidence first, then the calculation or rule, then the conclusion. If a value is absent, say what cannot be concluded. Keep any simulated mechanism distinct from a trained model or deployed service.

Teach the week as a story

The consent-safe poster: A team requests a synthetic poster. The output looks vivid but may show a recognisable person. How do generation and review interact?

Ask for a quick prediction before each technical explanation. Return to the prediction after the example and ask what evidence changed the answer. The analogies below are memory aids, and each has a stated limit.

Forward noising: analogy, mechanism and check

Familiar picture: A photograph becomes harder to see as fog builds in small steps.

Familiar part	Technical meaning
Clear image	training example
Added fog	forward noise
Fog level	time step

Where this analogy breaks: Noise is mathematically sampled; real fog is not the model's training process.

Actual idea: In a common training setup, progressively add noise to a clean image.

Work it aloud: At low noise the L-shaped grid remains visible; at high noise most cells are obscured.

Ask before revealing: Which direction is used while preparing training pairs?

Supported response: Forward: progressively add noise to a clean example.

Look for this misconception: The model learns by removing noise before any noise is added.

Denoising model: analogy, mechanism and check

Familiar picture: A restorer removes a little fog at a time after learning from many examples.

Familiar part	Technical meaning
Cloudy view	noisy sample

Estimated fog	model prediction
Clearer view	next step

Where this analogy breaks: A real model does not secretly have the exact clean target at generation time.

Actual idea: A learned model estimates how to remove noise at each step; the course simulation uses the known clean target.

Work it aloud: The prepared notebook's toy restorer uses a clean target to illustrate steps; label that shortcut.

Ask before revealing: What extra information makes the toy restoration easier?

Supported response: It has access to the original clean grid.

Look for this misconception: The toy denoiser proves real model image quality.

Conditioning: analogy, mechanism and check

Familiar picture: A poster brief steers an artist, but leaves room for choices.

Familiar part	Technical meaning
Brief	text condition
Artist decisions	denoising path
Draft poster	output

Where this analogy breaks: The brief does not guarantee that every requested detail appears correctly.

Actual idea: Text or other inputs can guide generation. The condition does not guarantee accurate details.

Work it aloud: Use "blue campus poster" for both trials; compare date lettering and layout separately.

Ask before revealing: What stays fixed in a fair output comparison?

Supported response: Keep the brief and checklist fixed; record changed generation settings.

Look for this misconception: A detailed prompt guarantees exact text in images.

Image review: analogy, mechanism and check

Familiar picture: A print editor checks a draft before it reaches the public.

Familiar part	Technical meaning
Draft	generated image
Release form	consent check
Proofread	image detail check

Where this analogy breaks: A release form cannot be inferred from pixels; ask for documented permission.

Actual idea: Check whether text, people, permissions and task-specific details meet the brief.

Work it aloud: The poster is attractive, but a face resembles a classmate and the date is wrong: hold publication.

Ask before revealing: What two reasons justify holding this image?

Supported response: Consent is unresolved and the date detail is incorrect.

Look for this misconception: A synthetic image needs no rights or quality review.

New situation for discussion

A poster has readable-looking text that spells the event date incorrectly. Which check catches the error?

Memory cue: Noise becomes an image through repeated learned denoising; check consent and details.

Guided laboratory

Run a denoising illustration and a consent checklist.

1. Inspect a 4 by 4 synthetic clean image
2. Add reproducible noise at two strengths
3. Use a transparent toy denoising demonstration
4. Compare two prepared descriptions using one rubric

Individual change: Increase noise strength from 0.15 to 0.45 and compare measured error. Expected evidence: Step comparison and three task-specific image observations.

Checks and common error

Measure: Mean absolute pixel error; plus a human quality checklist. Record the input, setting, source version, result and the relevant numerator and denominator.

Plausible incorrect claim: Removing random noise with the original image in hand reproduces diffusion generation.

Correction: This exercise visualizes steps only. Actual generation uses learned denoising without a clean target at inference.

If learners disagree, reproduce the same case and compare the underlying source or expected answer. Record unresolved cases for instructor review.

Further discussion

- How could the conclusion change if the source or input changes?
- Which part of the result is observed and which part is an assumption?
- What simple test would detect the failure described above?

References and optional extension

Core readings are official vendor documentation or original research. Read the relevant section; an account or paid API is unnecessary for this notebook. Page details may change before 2027.

Google, Develop a generative AI application.

<https://docs.cloud.google.com/docs/ai-ml/generative-ai/develop-generative-ai-application>

NVIDIA, NeMo framework overview.

<https://docs.nvidia.com/nemo-framework/user-guide/latest/overview.html>

Optional extension: replace one classroom proxy with a locally approved model or service, keep the original baseline, repeat the same checks, and state the additional privacy and cost assumptions.

Curriculum scope: the supplied 12-week teaching sequence, recorded in docs/SYLLABUS_ALIGNMENT.md. The original module descriptor is retained in sources/originals/07-Descriptor-Generative-AI-new-.docx. Supporting project notes: Deep-research.txt, RAG.txt, AI-Agents.txt, MCP-&-Vector-Database.txt, Multi-Agent-AI-System.txt, Eval-&-Fine-tune-&-Loop-Engineering.txt. These are planning sources, not empirical performance evidence.

Student revision and exam preparation

Outcomes: LO1, LO2 | 2 hours concepts + 2 hours guided practice

The idea in plain language

In a common diffusion training setup, a clean example is combined with noise. A model learns to predict noise or another denoising target. Generation starts from noise and applies learned reverse steps, often conditioned on text. Our toy knows the original image and therefore only illustrates the sequence.

A worked example

Compare two prepared poster outputs: A follows the brief but has unreadable lettering; B has clear lettering but depicts an unapproved real person. Both need revision for different reasons. Score adherence, legibility and consent separately.

Practice before reading the answers

1. Distinguish noising during training from denoising during generation. (4 suggested practice marks)
2. Does conditioning guarantee the requested text will be correct? (4 suggested practice marks)
3. Why is a toy that sees the clean image not a real diffusion generator? (4 suggested practice marks)

Draw the main diagram from memory, explain each arrow, then check it against the slides. Answer the three questions without notes before reading the model answers.

Model answers and marking guidance

1. Training creates corrupted examples and learns a reverse prediction. Generation applies learned reverse steps to a noise sample.
2. No. The condition guides generation; the resulting image still needs inspection.
3. It has access to the desired answer at inference. A trained generator must estimate reverse steps without that target.

For each four-mark practice answer: two marks for the correct idea, one for a relevant example or calculation, one for an appropriate explanation or limitation. These suggested practice marks do not change the module assessment scheme.

Official reading and video support

G4 Google Cloud Tech: Introduction to image generation

https://www.youtube.com/watch?v=kzzz8CO_oG4

Study focus: Denoising intuition. Video title and channel indexed in search; full transcript not retrieved.

N1 NVIDIA: Generative AI and LLM Learning Paths

<https://www.nvidia.com/en-us/learn/learning-path/generative-ai-llm/>

Study focus: Optional deeper RAG, diffusion and deployment training; some courses require payment.

Selected source concepts inform the lesson. Examples, figures, questions and solutions are original classroom material; no exam questions are claimed to be official.